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Automatic Detection of Demosaicing Image Artifacts and its Use in Tampering Detection

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***Abstract*—Even novice people can nowadays do convincing forged images. However, most forgeries alter the underlying statistics of the image, and in particular the slight artifacts caused by the demosaicing method. Demosaicing transforms the undersampled image acquired by the CFA into a three channel color image. Two problems arise: The first one is to identify the underlying CFA configuration, which classifies pixels according to whether they acquired a red, green or blue value. The second one is to detect anomalies in the regularity of the "demosaicing artifacts" that may reveal tampered image regions. We review the state of the art of detection methods, but point out that none of the proposed methods yields guaranteed detections for the CFA configuration or for the tampered areas. The methods generally yield an output that must still be evaluated visually. We therefore introduce an *a contrario* method that yields guaranteed detections, namely detections with a very low number of false alarms (NFA). Obtaining such an NFA is a useful complement to existing detection methods and should enable these methods to be included into automatic image evaluation processes.**

***Index Terms*—image forgery; forgery detection; forgery; CFA interpolation; CFA; color filter array ; demosaicing; demosaicking; filter estimation; linear estimation; a contrario; tampering; artifact detection; Bayer matrix**

I. Introduction

Intentional modifications of an image often leave trails by disturbing its underlying statistics, inherited from the image formation process. One of these statistics is caused by the Color Filter Array (CFA) interpolation, or demosaicing. Indeed, in most cameras, only one color value (R, G or B) is captured at each pixel, thus forming a mosaic image. The other two colors at each pixel must be interpolated from its neighbors by a demosaicing algorithm. The most frequent CFA is the Bayer matrix [1] which captures two green pixels, one red, and one blue pixel, see Fig. 1. The simplest possible demosaicing is the bilinear algorithm, which sets the missing pixels of a given color to the mean of its immediate neighbors of the same color. The bilinear CFA interpolation yields many artifacts, and the algorithms used by most cameras are far more complex, generally inspired from the nonlinear processes proposed in (e.g.) [2], [3], [4], [5], [6], [7], [8]. Regardless of the method, demosaicing leaves traces, as the interpolated pixels are more correlated with their neighbors than the observed ones. When an image is modified, these traces, or demosaicing artifacts, are easily altered. We are faced with a reverse engineering problem: to identify from a single image its CFA configuration, and even to estimate its demosaicing method. If done properly, this may open the way to detecting tampered regions, where a break in the demosaicing artifacts is sensed.

In this paper we address the problem of giving reliable statistical guarantees to the identification of the original CFA configuration (among the four possible ones). By localizing this CFA identification method to smaller blocks, we also show that, when the tampered regions are large enough, forgeries can also be detected with strong guarantees. Our review of the literature shows that, to the best of our knowledge, no current method has proposed such guarantees in the form of a NFA (number of false alarms). We now examine some of the prominent methods.

The seed paper on CFA-based tampering detection is [9]. Its detection methodology has been henceforth adopted by most papers. The core idea is that in the mosaiced image two classes of pixel colors exist: the observed ones acquired by the CFA and the interpolated ones. Making an assumption on the CFA configuration, one can simulate a demosaicing from the supposedly observed pixels, and see how well it reconstructs the supposedly interpolated pixels. If the assumed configuration is correct, the reconstruction should show an error significantly inferior to the other three configurations. In addition, examining locally the reconstruction error may reveal suspicious regions where this error is larger than elsewhere in the image. As shown in [9], a good detection can be obtained this way even with such a primitive demosaicing algorithm as bilinear interpolation. Then a spectral analysis of the residual error can detect a 2-periodicity for the residual error. This very same scheme was varied and improved in [10] and more recently in [11]. It is also adopted by [12] who use a PCA-based spectral analysis instead of Fourier, and by [13] who use the FFT and actually test several demosaicing methods and identify the most likely one by a trained neural network. This method was recently extended by the still more sophisticated algorithm in [14], where a more realistic generic anisotropic demosaicing method is estimated. This last method uses EM, in line with [9], to classify pixels according to their status (observed, interpolated). More recent methods estimate as accurately as possible the demosaicing method by introducing a parametric nonlinear method [15]. They even directly identify the used demosaicing method as done in the extended version [16].

As we mentioned, bilinear interpolation is the most popular

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$$\begin{matrix} R_{0,0} & G_{0,1} & R_{0,2} & G_{0,3} & \cdots \\ G_{1,0} & B_{1,1} & G_{1,2} & B_{1,3} & \cdots \\ R_{2,0} & G_{2,1} & R_{2,2} & G_{2,3} & \cdots \\ G_{3,0} & B_{3,1} & G_{3,2} & B_{3,3} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{matrix}$$

Fig. 1: Top left portion of a CFA image obtained with a Bayer matrix $\begin{smallmatrix} R & G \\ G & B \end{smallmatrix}$. At each pixel, a single color (either red, green, or blue) is sampled.

approximation of the demosaicing methods used for CFA identification. The method of [17] is an interesting variant: it applies a pseudo-inverse of this linear interpolation on each assumed acquired subgrid. This generates an error map depending on the CFA and the right CFA is again identified as the one with the minimal error. One method proposed by [18] uses again bilinear interpolation in a first version but proposes an interesting innovation: interpolated and acquired pixel grids are discriminated by denoising the mosaiced image (by a wavelet denoiser), and by detecting a significant difference for the noise variance between acquired and interpolated grids. The method is applied to complete images, but also on blocks with size 96 $\times$ 96. The rationale pointed out by several authors is that the variance of interpolated pixels should be significantly smaller than for the observed pixels [19]. This therefore provides a direct cue that avoids using a hypothetic demosaicing algorithm. A similar observation is made in [20] where the problem of identifying the CFA from an image is handled by examining the order of values rather than a reconstruction error. For example, to detect which of the two quincunx grids is the grid of acquired green pixels, the algorithm counts how many green values are between the min and max of neighboring values on each grid. The grid that has the higher count is detected as the grid of interpolated values. This method has also been applied in [21] to detect image regions where the color or hue have been manipulated.

The degree of sophistication of these methods seems quite sufficient. The main problem that now arises is that an image forensic examination requires many different tests, which in turn means a high risk of multiplication of false alarms. It also makes visual examination cumbersome if many images have to be examined.

In this context, it becomes particularly useful to control the rate at which false positives are found. The *a contrario* method [22] has been specifically developed in image analysis to that purpose. It considers a detection as meaningful if the expected number of similar detections obtained "just by chance" is below a specified false alarm rate. Here we aim at very low false alarms rates, which in practice offer full guarantees for strong detections.

To demonstrate the effectiveness of the method, we must first propose a simple and automatic estimate of the demosaicing method. From now on, we assume that a Bayer matrix (1) is used. There are four possible positions for the CFA grid: the first pixel can either be red, green on a line of reds, blue, or green on a line of blues. We denote these four possible

$$\begin{matrix} R & G & \cdots \\ G & B & \cdots \\ \vdots & \vdots & \ddots \end{matrix} \qquad \begin{matrix} G & R & \cdots \\ B & G & \cdots \\ \vdots & \vdots & \ddots \end{matrix} \qquad \begin{matrix} B & G & \cdots \\ G & R & \cdots \\ \vdots & \vdots & \ddots \end{matrix} \qquad \begin{matrix} G & B & \cdots \\ R & G & \cdots \\ \vdots & \vdots & \ddots \end{matrix}$$

RG/GB GR/BG BG/GR GB/RG

Fig. 2: The four possible CFA grid positions for a Bayer matrix.

positions respectively by $\begin{smallmatrix} R & G \\ G & B \end{smallmatrix}$, $\begin{smallmatrix} G & R \\ B & G \end{smallmatrix}$, $\begin{smallmatrix} B & G \\ G & R \end{smallmatrix}$, and $\begin{smallmatrix} G & B \\ R & G \end{smallmatrix}$ (Fig. 2). We do not know *a priori* which of these positions is used in the image: As such, we have no choice but to estimate the interpolation method for each of these assumptions. If the approximation is correct, the image obtained with the estimated method from the mosaiced image should be closest to the original when the good grid position assumption is made.

Considering a color image I of shape $(X, Y, 3)$, where $I[x, y, c]$ contains the value of the pixel at location (x, y) in the c^{th} channel – using the convention that the first channel is red, the second green and the third blue – we assume we know the position of the CFA grid G, of the same shape and with $G(x, y, c)$ equal to 1 if the c^{th} channel is the one sampled at position (x, y), and 0 otherwise. We construct the mosaicked image M of shape (X, Y) as

$$M[x, y] = \sum_{c=0}^{2} I[x, y, c] G[x, y, c].$$

If the grid position is correct, it corresponds to the image that was sampled by the camera.

Estimating the demosaicing method means we are looking for a function f such that f$(M) \approx I$. We shall approximate the demosaicing with linear filters as proposed by [9]. Our method is actually simpler because we shall avoid using an expectation maximization (EM) by estimating optimal filters for each of the hypothesized CFA configurations. Furthermore we get a better filter approximation by allowing them to use all observed channels, as is actually done in most modern demosaicing algorithms.

As we estimate a demosaicing method individually for each possible hypothesis on the grid position, we can directly find a least square optimal filter without recurring to EM. We can therefore directly find for each image the linear filter α of size $(2N + 1, 2N + 1)$ that minimizes

$$\sum_{c=0}^{2} \sum_{x,y} \left(I[x, y, c] - ((I \cdot G)[c] * \alpha)[x, y] \right)^2$$

where $*$ represents convolution and $\cdot$ the entry-wise product. In order to address the second problem and take inter-channel correlation into account, we have to use the mosaiced image M instead of a single channel of the original image masked on the interpolated pixels and minimize

$$\sum_{c=0}^{2} \sum_{x,y} \left(I[x, y, c] - (M * \alpha)[x, y] \right)^2 .$$

However, in order to solve the last problem and get an acceptable approximation, no less than 8 linear filters are to be estimated – one per position on the CFA grid/missing color channel pair:

- $\alpha_{R\to g}$ and $\alpha_{R\to b}$, which interpolate respectively the green and blue channels at the locations where the sampled pixel is red,
- $\alpha_{GR\to r}$ and $\alpha_{GR\to b}$, which interpolate respectively the red and blue channels at the locations where the sampled pixel is green on a line of reds,
- $\alpha_{B\to r}$ and $\alpha_{B\to g}$, which interpolate respectively the red and green channels at the locations where the sampled pixel is blue, and
- $\alpha_{GB\to r}$ and $\alpha_{GB\to b}$, which interpolate respectively the red and blue channels at the locations where the sampled pixel is green on a line of blues.

We estimate for instance $\alpha_{R\to g}$, the filter that interpolates the green pixel where the sampled pixel is red, by solving the linear systems that arise when we set to 0 the gradients of the L_2 error of the residual on pixels where the sampled channel is red. This yields the linear system $A\alpha_{R\to g} = b$, with

$$A[u+Nv, s+Nt] = \sum_{x,y} G[x,y,0]M[x+u,y+v]M[x+s,y+t]$$

and

$$b[u+Nv] = \sum_{x,y} G[x,y,0]M[x+u,y+v]I[x,y,1].$$

The other 7 filters can be estimated in a similar manner.

For each grid position $P \in \{{}^{R\,G}_{G\,B}, {}^{G\,R}_{B\,G}, {}^{B\,G}_{G\,R}, {}^{G\,B}_{R\,G}\}$, let $\tilde{I}_P$ be the image interpolated from the corresponding mosaicked image with the filters estimated above. Notice that the same linear filters can be evaluated locally by dividing the image into windows with sufficient size.

II. Estimation of a Meaningful Grid Position

At this step, we want to know which grid position is used in the image, as well as in image sub-windows. The easiest way would be to check which residuals have the lowest norm. However, we also need to be sure that this grid position is significantly more likely than the other, since there are a variety of reasons for which no CFA interpolation traces may be detected – the image could have been falsified, we could be in a flat sub-window, or the interpolation could be undetectable either because of a high compression or as a result of sub-sampling.

We are looking for an *a contrario* approach [22] so as to control the expected number of times where a false grid will be detected.

We subdivide the image in squares blocks of size $b\times b$, with b even so as to fully contain all of its unit Bayer matrices, and we require each block to vote for the grid position whose residual in the block is smallest. Notice that this step can be done with most of the detection methods we reviewed, as they can provide such a local vote.

Let $P \in \{{}^{R\,G}_{G\,B}, {}^{G\,R}_{B\,G}, {}^{B\,G}_{G\,R}, {}^{G\,B}_{R\,G}\}$ be the position that was most voted by blocks. We want to decide whether this relative excess of votes is significant. Let n be the number of blocks in the image or window, and n_P be the number of those blocks that voted for position P. We want to control the number of false alarms (NFA) [22], that is, the expected rate at which this grid position would be voted "just by chance" by as many blocks or more, in an image where no demosaicing was actually applied. In such an image, the grid choice for each block is purely random and each configuration has the same probability.

Thus, our *a contrario* stochastic model is an image where the vote for the CFA configuration is a white noise over a set of disjoint blocks. This noise has four possible values, each with probability $1/4$. Let now p_g be our intended number of false alarms. A value of $p_g = 0.001$ means for instance that on average we would observe one window with so many votes for a same configuration in one in 1000 images respecting the *a contrario* model.

In this model, the probability for a block to vote for a given position is $\frac{1}{4}$, and the probability that there are at least n_P blocks voting for this given position therefore is, by the binomial law,

$$\sum_{i=n_P}^{n} \binom{n}{i} \left(\frac{1}{4}\right)^i \left(\frac{3}{4}\right)^{n-i}.$$

As there are 4 possible configurations, and since the image is subdivided into z disjoint windows in which we work individually ($z = 1$ if we are working on the whole image), the expected number of false alarms, or NFA, is

$$\mathrm{NFA}(n_p, n) = 4z \cdot \sum_{i=n_P}^{n} \binom{n}{i} \left(\frac{1}{4}\right)^i \left(\frac{3}{4}\right)^{n-i}.$$

Thus, by considering a CFA configuration P meaningful only if $\mathrm{NFA}(n_p, n) \leq p_g$, we ensure that the expected number of false alarms will be p_g [22].

III. Automatic Forgery Detection

Falsifications may alter the CFA interpolation traces in two ways:

- they can locally destroy all interpolation traces, or
- when part of an image is pasted from somewhere else (either in this image or from another image), they can locally introduce another grid position.

As traces of interpolation may be naturally absent, mostly due to compression, it is not possible to know whether the lack of interpolation traces is due to forgeries just by using the votes of blocks. Such a problem is thus out of the scope of this article and we focus on the second kind of forgeries. One should also note that, if part of an image is pasted onto another, thus bringing new CFA interpolation traces, there is one chance out of four that the grids positions will be the same, in which case no CFA based method will not be able to detect the forgery. As we can only find localized forgeries if traces of demosaicing in the global image are significant, we only consider this case. Let P_0 denote the main position of the CFA grid in the global image.

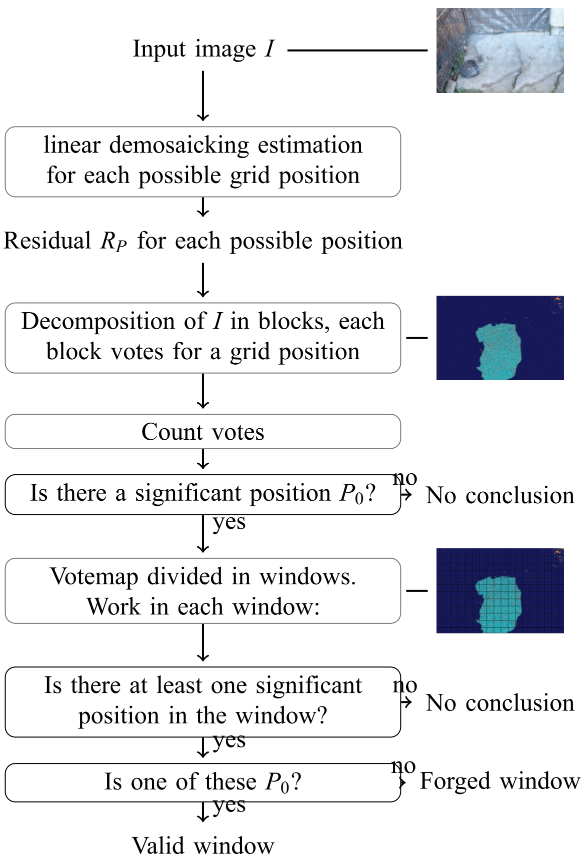


Fig. 3: Global workflow of our method

As before, we want to control the number of false alarms, that is, the rate at which a window will wrongly be considered as tampered with. Let p denote the expected number of windows per image that will be declared as falsified under the *a contrario* assumption that the image was not mosaiced. So as to keep control over the number of false alarms, a position P can be considered significant in window W up to NFA threshold p if $\mathrm{NFA}(n_{W_P}, n_W) \leq p$. We consider a window to be forged if there is at least one significant position in it, but none of them are P_0.

IV. Results

A. Approximation of Demosaicking Methods

We took images from the Kodak dataset [23] and demosaiced them manually with several methods. We then used our algorithm to approximate the interpolation method, reinterpolated them with the estimated filters, and compared the result to the original image, see Fig. 4. The bilinear demosaicking, by nature linear, was estimated perfectly by our method, even when the image was corrupted by high JPEG noise.

These results show that, while having only one linear filter cannot approximate correctly even the simplest methods, estimating one different filter for each possible pair sampled pixel – interpolated channel, for a total of eight filters, yields much better results.

	BI	CS
5×5	0.–0.0180	0.0044–0.0049
7×7	0.–0.0127	0.0034–0.0041
$F7\times7$	0.2842–0.2848	0.2322–0.2322
	GT	ZW
5×5	0.0036–0.0039	0.0042–0.0050
7×7	0.0021–0.0030	0.0027–0.0039
$F7\times7$	0.2366–0.2366	0.2322–0.2322

Fig. 4: Average RMSE on the images of the Kodak dataset, demosaiced respectively with bilinear interpolation (BI), contour stencils (CS) [6], Gunturk (GT) [7] and Zhang-Wu (ZW) [8], and reinterpolated with estimated filters of size 3×3 and 7×7. In each cell, the first value represents the RMSE when the estimation is done on the correct grid, and the second value represents the mean of the three RMSE obtained when the estimation is done on one of the three other grids. A large difference between the two values means that the correlations caused by demosaicing were correctly captured in the filter estimation, and enables us to distinguish the correct grid position more easily. For comparison, the values in the row $F7\times7$ correspond to the use of one filter per channel, without using interchannel correlation, as done in [9].

(a) 28 blocks of size 127×127, 82% of votes were for the position $\begin{smallmatrix}R&G\\G&B\end{smallmatrix}$, $\mathrm{NFA}=1.4\times10^{-9}$.

(b) 384 blocks of size 32×32, 78% of the votes were for the position $\begin{smallmatrix}R&G\\G&B\end{smallmatrix}$, $\mathrm{NFA}=5.7\times10^{-27}$.

Fig. 5: Votes of blocks of different sizes on a remosaiced image of the Kodak dataset.

B. Estimation of the Grid Position

Once the filters have been approximated separately for the four possible grid positions, we decompose the image in blocks of different sizes and estimate the most significant grid position and its NFA, see Fig. 5

As could be expected, the NFA is lower when there are more blocks, which means we should rather use blocks of size 2×2. The NFA was below the machine precision of 10^{-324} when we had blocks of size 8×8 or lower.

We also checked robustness of the grid detection to JPEG compression, see Fig. 6. Although the increasing NFA for lower qualities means we cannot hope to detect small forgeries with strong certainty, we can see that the traces of CFA interpolation are still visible enough in high qualities.

C. Detections in Noise

In order to validate our *a contrario* model, we took 800000 white noise images of size 32×32 and tried to detect a

Quality	Detection percentage	NFA
100	100	10^{-300}
99	100	10^{-40}
97	98	10^{-10}
95	81	10^{-6}
90	67	10^{-4}

Fig. 6: Robustness to JPEG Compression on the 24 images of the Kodak database, reinterpolated with contour stencils[6], and compressed with various qualities. We show the percentage of images for which a significant grid position was detected, and the mean NFA associated to detected grids. When a significant grid position was detected, it was always the correct one.

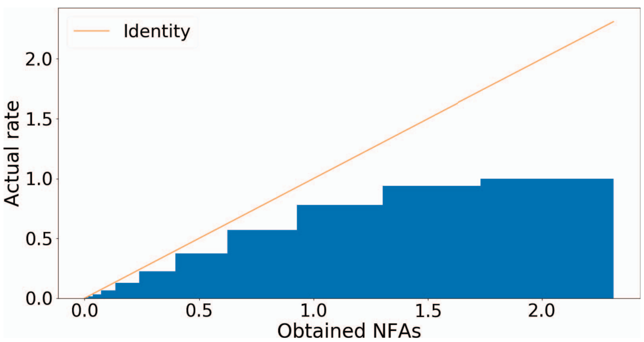
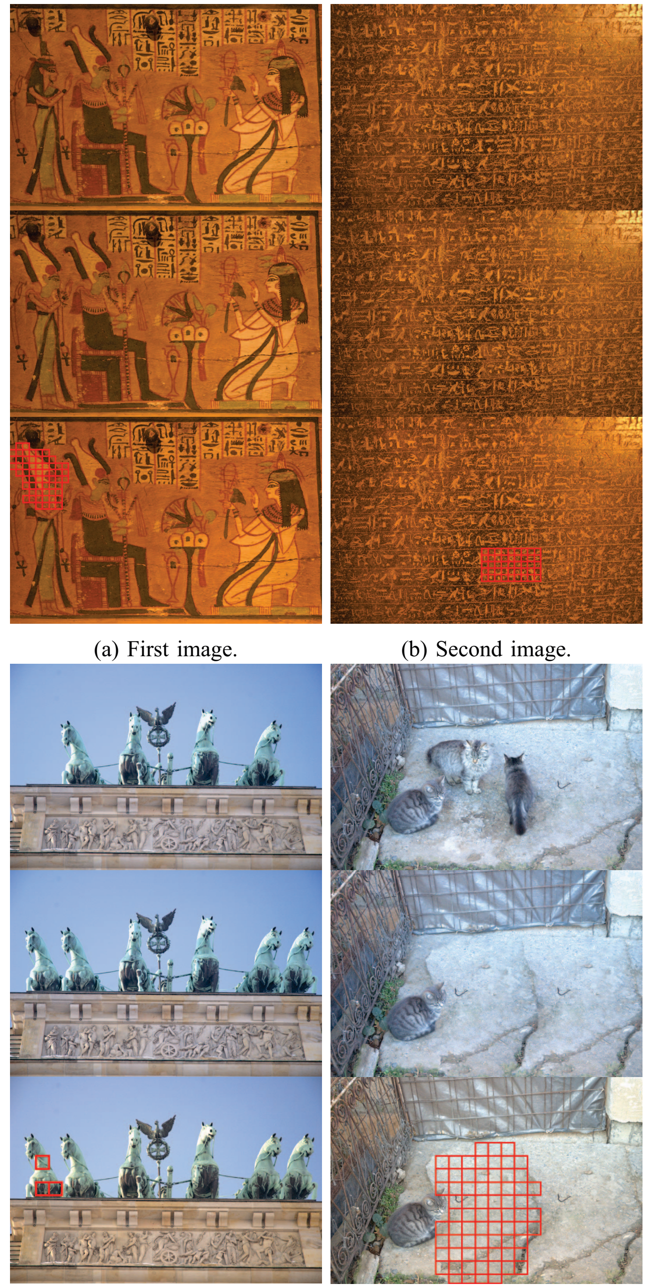


Fig. 7: Cumulative histogram of the obtained NFAs in noise. The x axis corresponds to the theoretical NFA that our algorithm outputs, it represents the theoretical rate at which a result at least as significant would be wrongly obtained in noise, while the y axis represents the actual rate at which we obtain such a rate in noise ; For instance, the 1.0 value at the 2.0 bar means that, in noise, our algorithm wrongly detect once per image a value for which one should get more than twice per image. As such, our model can be validated as long as the histogram is below the identity line.

significant grid position, see Fig. 7. We clearly see that the rate of false alarms is always lower than indicated by the algorithm, which means that, in noise, we can expect a true number of false alarms that is at worst as high as the output of the algorithm.

D. Forgery Detection

We took a few tampered images from [24] and analysed them with our algorithm. We set the NFA threshold at 10^{-10} in order to ensure that there would be practically no false alarms. We could not detect most of the smaller forgeries, mainly because they required working in smaller windows, in which it was thus harder to achieve significance below the NFA threshold. See Fig. 8 for examples of detected forgeries. It is interesting to see that in these images, the windows detected as forged did all have a NFA lower than 10^{-100}, despite the relatively small window size of 32×32 blocks of 4×4 pixels.



(a) First image.

(b) Second image.

(c) Third image. Only one of the two added horses was detected, probably due to the fact that the right horse was pasted at the correct position on the CFA grid, which can happen with probability $\frac{1}{4}$.

(d) Fourth image.

Fig. 8: Original images from [24], their corresponding forgeries and our detection of those forgeries.

One should note that if the global grid position is highly significant, the proportion of blocks voting for this position is higher, which in turns means that we can detect significant positions with smaller windows. Further work could establish theoretically how large windows would need to be to achieve signifiance, depending on the global significance of the correct grid position.

V. Discussion

In this article, we constructed a method allowing us to correctly approximate demosaicing methods with several linear filters. We then used an *a contrario* model, not only to detect which grid position was the most likely, but also how significant it was, by keeping under control the number of false errors. Still within an *a contrario* model, we then tested windows of the image to check which of these were likely to have been forged.

By working separately on the four possible Bayer matrix positions, we saw that an optimal linear approximation of demosaicing algorithms was efficient enough. This may be explained by the fact that the highly non-linear nature of most demosaicing algorithms can partially be balanced by estimating filters in local areas and not for the whole image. We have verified that the demosaicing artifacts could still be detected under JPEG compression, as long as the compression quality was high enough.

We also proposed an *a contrario* approach to the identification of falsified parts of an image that voted for a different CFA configuration. Clearly such a method risks a false negative rate of 1/4, as it might happen just by chance that a copy-paste operation respected the CFA configuration. Yet when detected, controlling the number of false alarms could prove highly useful in image forgery detection, where a battery of complementary tests is often necessary given the variety of possible falsifications.

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